OMNITOME PUBLISHING | SYNTHETIC ARCHIVE

The Art and Science of Reading

OCR Test Plates | October 2026



Oculomotor Psychophysics and the Physiological Speed Ceiling

1. Retinal Acuity & Fixations

Human reading speed is governed by the steep anatomical drop-off of cone photoreceptors outside the central retina. High-acuity letter discrimination occurs exclusively within the central 2° fovea, spanning 6 to 8 character spaces at normal reading distance. Surrounding the fovea from 2° to 5° lies the parafovea, which extracts word-length spacing and initial-letter preview benefits across 14 to 15 characters to the right of fixation in English (and 14 characters to the left in Hebrew).

During skilled reading, the eyes pause in stationary fixations averaging 225 to 250 milliseconds, separated by ballistic saccades lasting 20 to 35 milliseconds and spanning 7 to 9 character spaces. Readers directly fixate approximately 85% of content words and 35% of function words, while 10% to 15% of saccades are backward regressions required to resolve syntactic ambiguities.

2. The 500-WPM Retinocortical Limit

Commercial speed-reading courses claiming 1,000 to 3,000 words per minute (WPM) violate retinocortical transmission latencies (Rayner et al., 2016). Visual transmission from the retina to the left Visual Word Form Area (VWFA) requires 50 to 60 ms, lexical familiarity verification (Stage L1) requires 65 to 100 ms, and overlapping saccade motor programming sets a minimum sustainable fixation floor of 190 ms plus 25 ms of saccadic flight (215 ms per cycle, or 279 fixations per minute).

Because foveal and parafoveal acuity can identify at most 1.5 to 1.8 words per fixation even with zero regressions, multiplying 279 fixations/min by 1.8 words/fixation establishes a hard physiological ceiling of approximately 502 WPM. Any reading rate above 600 WPM represents macrostructural skimming rather than full-text comprehension.

MARGINAL NOTE

Screen Deficit Moderator: Across 54 studies (*N* = 171,055; Delgado et al., 2018), screens impair comprehension on informational texts (*g* = −0.25) and under time limits (*g* = −0.26) due to metacognitive overconfidence (Ackerman & Goldsmith, 2011).

The Testing Effect Crossover (Roediger & Karpicke, 2006)

While repeated rereading (SSSS) produces higher immediate 5-minute recall than retrieval practice (STTT) by elevating perceptual fluency in the Visual Word Form Area, active retrieval practice dramatically reduces 1-week forgetting.

Study Condition	Mean Exposures	5-Minute Recall	1-Week Delayed Recall	1-Week Forgetting
Repeated Study (SSSS)	14.2 readings	81%	42%	52% forgotten
Single Test (SSST)	10.3 readings + 1 test	75%	56%	25% forgotten
Retrieval Practice (STTT)	3.4 readings + 3 tests	70%	**61%**	**13% forgotten**

The Omnitome Two-Representation Invariant: Pairs normalized 1,200 to 2,400-word Original Reading View segments with 250 to 350-word provenance-linked Smart Chapters indexed by 1024-dimensional `BGE-M3` embeddings (`[CLS]` pooling calibration improving retrieval separation from **0.2520** to **0.4225**).